When LLMs Go Abroad:
Foreign Bias in AI Financial Predictions

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Abstract

We document "foreign bias" in AI financial predictions, reversing the classic home bias. U.S.-based ChatGPT is systematically more optimistic than China-based DeepSeek about Chinese firms—in price predictions and directional forecasts—yet significantly less accurate. Evidence supports an information-availability mechanism: bias is strongest when U.S. media coverage of Chinese firms is limited and attenuates for cross-listed firms. Crucially, injecting Chinese news eliminates the prediction gap. Both models produce similar forecasts for U.S. firms, consistent with broader worldwide coverage. LLMs trained in different information environments can create divergent signals, with implications for investors and policymakers as AI increasingly intermediates global markets.

Keywords: Artificial intelligence; Financial statement analysis; Large language models; ChatGPT; DeepSeek

JEL: D83, L86, G14, G15, G41, M41, O33

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1 Introduction

The rapid adoption of Large Language Models (LLMs) in financial analysis represents a fundamental shift in how investment information is produced and consumed (Engelberg, Manela, Mullins, and Vulicevic, 2025; He, Lv, Manela, and Wu, 2025). ChatGPT reached 100 million users within two months of its November 2022 launch, with surveys indicating that over 40% of institutional investors now use AI tools for market analysis (Bloomberg Intelligence, 2024); other market participants also appear to be integrating the tool into their workflows (Blankespoor, DeHaan, and Li, 2025; Cheng, Lin, and Zhao, 2025; Law and Shen, 2025). A nascent literature documents LLMs' ability to process or synthesize financial statements (Bertomeu, Lin, Liu, and Ni, 2025; Bradshaw, Ma, Yost, and Zou, 2025; de Kok, 2025; Shaffer and Wang, 2024), and to forecast earnings (Cook, Kazinnik, Hansen, and McAdam, 2023; Kim, Muhn, and Nikolaev, 2024; Kim and Nikolaev, 2025), corporate policies (Jha, Qian, Weber, and Yang, 2024), and stock returns (Lopez-Lira and Tang, 2023; Chen, Green, Gulen, and Zhou, 2024).

However, a growing body of work in finance reveals potential perils in applying LLMs to financial analysis. One stream focuses on *function-centric* limitations, documenting extrapolation errors, miscalibration (Chen et al., 2024), and behavioral anomalies that can distort predictions (Bini, Cong, Huang, and Lawrence, 2025). A second stream studies *information-set* distortions, most notably temporal distortions that lead to look-ahead bias and leakage (Levy, 2025; Sarkar and Vafa, 2024; He et al., 2025). We extend this second stream by studying a novel information-set distortion: *spatial* (geographic) gaps in training corpora arising from cross-country differences in news coverage. As investors increasingly rely on LLMs trained in different countries—and potentially shaped by very different information environments—understanding whether and how these spatial gaps affect financial predictions becomes essential.

The parallel development of frontier models in the US and China provides a unique opportunity to address this question. Cross-border model comparisons offer three key advantages. First, when two frontier models trained in different environments analyze identical firms, systematic prediction differences provide insights into potential sources of bias. Second, this design controls for firm characteristics and allows for attribution of prediction differences to model-level factors, such as how models process information or how their predictions are affected by asymmetries in their underlying training data, rather than to cross-sectional variation in firm fundamentals. Third, as the United States and China emerge as the two leading AI powers, understanding how their leading models differ has direct implications for both practitioners and academics. These differences can shape cross-border investment analysis, market efficiency, and the global diffusion of AI.

Our investigation draws parallels between human and algorithmic decision-making under information asymmetries. While humans exhibit home bias—overweighting domestic or local assets despite diversification benefits (French and Poterba, 1991; Cooper and Kaplanis, 1994; Tesar and Werner, 1995; Lewis, 1999; Coval and Moskowitz, 1999)—the mechanism remains debated: information advantages (Portes and Rey, 2005; Van Nieuwerburgh and Veldkamp, 2009) versus behavioral biases that lead investors to favor familiar assets (Heath and Tversky, 1991; Huberman, 2001). Whether LLMs exhibit similar patterns is an open and compelling question because they learn from human-generated text that could embody human biases (Blodgett, Barocas, Daumé, and Wallach, 2020; Bender, Gebru, McMillan-Major, and Shmitchell, 2021; Shumailov, Shumaylov, Zhao, Gal, Papernot, and Anderson, 2023), or could produce new biases due to how information is incorporated.

We prompt both ChatGPT 4.1 and DeepSeek R1, frontier models from the US and China, respectively, to provide six-month price predictions and directional forecasts for nearly 5,000 firms listed on China's Shanghai and Shenzhen exchanges as of June 30, 2024. This design allows us to observe how two different models assess identical firms at the same point in time. Both models share a common reported knowledge cutoff of June 30, 2024, ensuring our December 31, 2024 target date represents a genuine six-month out-of-sample forecast for both.

We document a striking reversal of traditional home bias when LLMs analyze foreign firms. While humans favor familiar domestic assets, we find ChatGPT (US-based) is systematically more optimistic than DeepSeek (China-based) about Chinese firms, exhibiting what we term "foreign bias." Predictions generated by ChatGPT are 2.415 RMB higher than DeepSeek, representing 12.53% of the average and 19.68% of the median stock price, and directional forecasts generated by ChatGPT are 1.3 percentage points more likely to be "buy." This foreign bias is not explained by firm characteristics—the differences persist after controlling for size, profitability, leverage, past returns, and other observables that might influence AI assessments. Crucially, we confirm that ChatGPT's forecasts reflect an optimism bias rather than superior information: they exhibit 13.3% larger absolute price prediction errors and 7% higher directional forecast error rates than DeepSeek's predictions when compared to realized December 31, 2024 outcomes.

What explains ChatGPT's foreign bias? We consider two broad channels: information-centric and function-centric.[1] Under an information-centric explanation, relevant information about Chinese firms may be sparse or absent from ChatGPT's training information set (data availability). Under function-centric explanations, such information may be present but weakly encoded in the learned parameters (parameter encoding), or it may be encoded but not reliably retrieved or applied under our prompting regime (elicitation). These channels yield different testable implications. If missing information is the primary driver, we should expect ChatGPT's relative optimism to be smallest for Chinese firms where cross-market information gaps are smallest; moreover, we should expect that providing it Chinese news via prompts should eliminate the bias. If instead information is present but underweighted, injected news must compete with existing priors and the bias should attenuate but may persist. If elicitation differences are central, the bias should vary with prompt language, format, or model variant.

Our evidence points to information availability as a likely driver of this foreign bias. Using comprehensive news data from RavenPack (US media) and the China Finance News Database

[1]We view an LLM as implementing a conditional distribution $p_{\theta_{\mathcal{I}}}(\cdot \mid x)$ over outputs given a prompt x, where $\theta_{\mathcal{I}}$ denotes parameters learned from a training information set $\mathcal{I}$. Cross-model differences can therefore arise from (i) differences in the information available during training ($\mathcal{I}$) and (ii) differences in how that information is represented in the learned parameters ($\theta_{\mathcal{I}}$) and elicited at inference (i.e., how the prompting/generation protocol maps a given context x into outputs).

(Chinese media), we construct firm-level measures of cross-market news coverage gaps. Chinese outlets account for the vast majority of news for many Chinese firms, and a large fraction of them receive little or no US coverage. Consistent with an information-availability mechanism, ChatGPT's relative optimism is most pronounced at firms where US media coverage, especially negative news coverage, is relatively scarce. Moreover, when we control for these firm-level news gaps, the baseline foreign-bias estimate becomes statistically and economically insignificant, implying no differences in ChatGPT's predictions in the absence of cross-border information gaps, consistent with the information availability hypothesis.

Cross-listed firms provide another natural test of this mechanism. Chinese firms listed on US exchanges receive substantially more English-language disclosure and attention (e.g., SEC filings, analyst reports, and US financial press), narrowing the cross-market information gap. Consistent with the information availability channel, we find that ChatGPT's relative optimism is significantly attenuated for cross-listed firms: the foreign bias is reduced by 30% for price predictions and 60% for directional forecasts.

Our most direct evidence comes from manipulating the LLMs' information sets available at inference time. We augment the prompt for both ChatGPT and DeepSeek with all Chinese-language firm news from June 2022 to June 2024.[2] When we provide ChatGPT and DeepSeek with Chinese news articles about each firm before requesting predictions, the US-model's excess optimism becomes economically and statistically insignificant. This intervention helps distinguish between information availability and parameter encoding explanations: prompting can add firm-specific information to the model's context but cannot change the underlying parameters learned during training. If the bias reflected Chinese information being present but systematically underweighted in the model, injected news would compete with existing priors and would be expected to only partially attenuate the foreign bias. Instead, we find that information provision alone eliminates

[2]Because providing the full text of all articles exceeds practical limits, we supply the abstracts for all available Chinese news articles in that window (rather than full text). However, in an alternative implementation, we instead provide five randomly sampled full-text Chinese news articles from the same window, and find very similar in the results.

it: the foreign-bias coefficient becomes -0.798 (insignificant) for prices and -0.002 (insignificant) for directional forecasts, compared to baseline differences of 2.415 and 0.013 respectively. Moreover, we show that this news-augmentation does not impact DeepSeek's output yet dampens the optimism of ChatGPT's. This pattern is consistent with ChatGPT's relative optimism stemming from missing information rather than from weak encoding of available information.

Moreover, the information availability mechanism predicts that foreign bias need not be symmetric. If English-language information about US firms is abundant and comparably available to both models, while Chinese-language information is more extensively covered in DeepSeek's training data, then the informational gap should be larger when the LLMs evaluate Chinese firms than US firms.[3] Consistent with this asymmetry, when both models analyze US firms, prediction differences are small and statistically insignificant, consistent with comparable information access in that setting.

While these results align with an information-availability interpretation, they could also reflect differences in how the models respond to prompts ("elicitation differences"), another function-centric difference between LLMs. We therefore conduct placebo and robustness tests designed to assess the scope for elicitation-based explanations. First, we ask both models to make predictions about synthetic (nonexistent) firms with Chinese-sounding names. By design, there is no underlying firm-specific information for either model to differentially access; any systematic bias would therefore be consistent with name-triggered elicitation rather than information availability. We find no prediction differences in this placebo setting. Second, we re-run the prompts in Chinese and find that the bias persists, suggesting it is not driven by an English-only prompt framing. Third, we test whether inference-time scaffolding changes the magnitude of the bias. Holding prompt template fixed, enabling online search—which expands the information available to the model at

[3]This asymmetry could be accentuated if DeepSeek has broad access to English-language information—potentially including via exposure to outputs from leading English models, as has been alleged in public reporting—while additionally drawing on richer Chinese-language sources. See Nellis et al. (Reuters, Jan. 29, 2025) and Weatherbed (The Verge, Jan. 29, 2025) for reporting on OpenAI/Microsoft concerns about possible "distillation" / unauthorized use; the claim was also reported by the Financial Times (Jan. 28, 2025).

prediction time—attenuates, but does not eliminate, the foreign bias. This result is consistent with missing or hard-to-retrieve information contributing to the baseline gap, while suggesting that ad hoc retrieval does not fully close the models' effective information gap. Moreover, to the extent reasoning-oriented models reduce reliance on superficial prompting cues, a purely elicitation-driven mechanism would predict attenuation. We show that the bias persists when replicating the analysis using ChatGPT-5, a reasoning-oriented model variant. Finally, we show that the bias persists at a longer forecast horizon (12 months), suggesting it is not confined to short-horizon framing. Taken together, these tests narrow the scope for elicitation-based explanations and indicate that the foreign-bias pattern is stable across language, identity-cue placebos, retrieval access, model variants, and forecast horizon framing, and responsive in the expected direction to exogenous increases in available information (web search).

Taken together, the coverage-gap heterogeneity, cross-listing attenuation, news-injection results, and the one-sided bias all point to *effective* information availability/access as a first-order driver of the AI foreign-bias we document. While we cannot directly observe training corpora or parameter-level representations, the weight of evidence favors an information-centric interpretation over function-centric explanations.

This paper makes three main contributions to the literature. First, we document a novel LLM phenomenon—*foreign bias*—in which (US-based) ChatGPT is systematically more optimistic than (China-based) DeepSeek about Chinese-firm stocks and less accurate ex post. This pattern reverses the classic "home bias" documented for human investors, underscoring that behavioral finance intuitions do not automatically transfer to LLMs.

Second, we provide evidence that this foreign bias is most consistent with *spatial information gaps* across countries—exogenous differences in the informational environments from which models learn. Whereas much of the AI-bias literature emphasizes endogenous sources (e.g., developer choices, alignment, prompt design, or temporal leakage), our findings highlight that model biases can also arise from *exogenous* cross-market information frictions.

Third, we offer a cross-border perspective on LLMs as information intermediaries. By comparing ChatGPT and DeepSeek—frontier models developed in different jurisdictions—we show that the same firm can receive systematically different assessments depending on which jurisdiction's model is used. This extends the information asymmetry literature to a new intermediary: unlike analysts or media, who can invest in information acquisition to mitigate asymmetries (Brennan and Subrahmanyam, 1995; Bushee, Core, Guay, and Hamm, 2010), LLMs can inherit and amplify existing cross-border information gaps through their training and inference behavior.

The resulting divergence in LLM forecasts has implications for cross-border investment analysis, market efficiency, and for policy discussions around transparency in model training data and sourcing. An immediate practical implication is that when investors or analysts use LLMs for cross-border financial analysis and prediction tasks, certain models' outputs can be systematically skewed, potentially distorting investment inferences and risk assessments, and mitigating these distortions may call for explicit local-information supplementation (e.g., local-language sources) and validation against alternative models or benchmarks. More broadly, our findings suggest that unknown and exogenous information environment differences can result in unexpected biases in LLM predictions. As AI tools increasingly mediate cross-border capital allocation, understanding and monitoring these biases will be important for market efficiency and investor protection.

2 Sample Construction and Data

2.1 Sample Selection

Our sample construction begins with the universe of Chinese firms listed on the Shanghai and Shenzhen stock exchanges with available data in Compustat Global. We focus on Chinese firms as they represent a major cross-border investment opportunity where information asymmetries between US and Chinese investors are particularly pronounced. The parallel development of sophisticated LLMs in both countries—ChatGPT in the US and DeepSeek in China—provides an ideal setting to study how AI models trained in different linguistic and informational environments assess the same set of firms.

We start with all Chinese firms in Compustat Global as of December 31, 2023, requiring non-missing data for key financial variables used as controls in our analysis. Specifically, we require firms to have available information on total assets (to calculate $FIRMSIZE$), return on assets (ROA), and leverage (LEV). These screens ensure that both AI models have access to basic financial information when making predictions, allowing us to isolate the effect of information asymmetries in news and contextual information rather than differences in fundamental financial data availability. This process yields 4,978 unique Chinese firms.

2.2 AI Prediction Collection

For each firm in our sample, we collect stock price predictions and directional forecasts (up or down) from two frontier LLMs: ChatGPT 4.1 (developed by OpenAI) and DeepSeek R1 (developed by DeepSeek). We access both models through their respective APIs to ensure consistency and replicability. It should be noted that both ChatGPT4.1 and DeepSeek R1's training sample periods ended around June and July, 2024, with all predictions made for the same future date of December 31, 2024, creating a almost six-month out-of-sample prediction horizon.[4] This can mitigate the influence of the forward-looking bias of LLM on our results (Didisheim, Fraschini, and Somoza, 2025).

We use standardized prompts to ensure comparability across models. For price predictions and directional forecasts, we prompt each model with: "Here is a firm called "name" (ticker: ticker), listed as an A-share in China. As a professional financial analyst, please: 1. Predict its stock price on 31 December 2024 (three decimal places). 2. Predict whether the price will increase from

[4]The cutoff dates reflect the vendors' public statements regarding each model's training data, with June 30, 2024 used for ChatGPT-4.1 and July 31, 2024 for DeepSeek R1. While ChatGPT and DeepSeek differ by one month in their reported training cutoff dates, this discrepancy is unlikely to affect our findings. Our U.S. firm tests yield no systematic differences across models, suggesting that minor differences in training horizons do not materially influence the results.

the end of June this year to December 31, 2024 (1 = up, 0 = not up). Do NOT use current online data or random numbers - use your best effort." We randomize the order in which firms are presented to each model to avoid any potential ordering effects. To ensure data replicability, we set the temperature to zero. To ensure data quality, we re-query both models for all our sample firms one week later to verify consistency, finding correlation coefficients exceeding 0.95 for both models' predictions. (Exhibit A reports the prompts used across all analyses reported in this paper, including the baseline specification and each robustness test.)

This process generates 9,956 firm-AI observations (4,978 firms $\times$ 2 AIs), with each observation containing a price prediction (*AI_PREDICTION*) and a directional forecast indicator (*AI_RECOMMEND*, coded as 1 for expected increase in stock price and 0 otherwise).

2.3 News Coverage Data

To examine the mechanism driving AI prediction differences, we collect comprehensive data on media coverage from both US and Chinese sources. This allows us to infer the role of cross-border asymmetries in news coverage about Chinese firms in explaining the observed foreign bias. Our hypothesis is that ChatGPT's training prioritizes US sources, whereas local Chinese news plays a greater role in DeepSeek.

For US media coverage, we use RavenPack News Analytics, which processes news from major US financial publications including the Wall Street Journal, Financial Times, Bloomberg, Reuters, and Dow Jones Newswires (Drake, Guest, and Twedt, 2014; Guest, 2021; Lock, 2024). RavenPack provides sentiment scores for each news article mentioning our sample firms. We aggregate all news articles from June 30, 2022, to June 30, 2024 (a three-year window) to capture the information environment likely reflected in the LLMs' training data. Articles are classified as negative if they have a sentiment score below 0 on RavenPack's [-1, +1] scale, and positive if above 0.

For Chinese media coverage, we use the China Finance News Database (CFND), which aggregates content from major Chinese financial media outlets including Caixin, Securities Times, China Securities Journal, and Shanghai Securities News (Li, Li, Wang, and Thatcher, 2021; Gao, Zhang, and Yang, 2023; Qiao, 2023). Similar to our approach with RavenPack, we collect all articles mentioning our sample firms during the same three-year period and classify them by sentiment using CFND's proprietary sentiment analysis, which is calibrated to be comparable to RavenPack's methodology.

For both sources, we impose a minimum article length requirement of at least 10 sentences. This criterion ensures that we eliminate those articles with minimal firm-specific information, such as brief wire alerts and index-inclusion notices, and focus on more substantive articles. To the extent these media sources are used in the training of LLMs, such short items are unlikely to meaningfully shape an LLM's learned representations of a firm.

We then construct two firm-level measures of cross-border information gaps. *NEG_NEWS_GAP* is defined as

$$\frac{\textit{CN negative} - \textit{US negative}}{\textit{CN negative} + \textit{US negative}} \tag{1}$$

computed over the three-year window prior to the models' training cutoff (June 30, 2022 to June 30, 2024). Here, *CN negative* refers to the number of negative news articles about a given firm from Chinese media sources (CFND), while *US negative* refers to the number of negative news articles from U.S. media sources (RavenPack). This variable lies in $[-1, 1]$, with $+1$ indicating only Chinese negative coverage and -1 indicating only U.S. negative coverage.

NET_NEG_NEWS_GAP is defined as

$$\frac{(\textit{CN negative} - \textit{CN positive}) - (\textit{US negative} - \textit{US positive})}{\textit{CN negative} + \textit{CN positive} + \textit{US negative} + \textit{US positive}} \tag{2}$$

where *CN positive* (*US positive*) denotes the number of positive news articles about the firm from Chinese (U.S.) media sources. This measure captures the net negativity gap across countries, with higher values indicating that Chinese outlets are, on net, more negative than U.S. outlets. For firms with zero articles in both markets (denominator $= 0$), we set the gap to zero; results are robust to dropping such firms or adding a no-news indicator.

2.4 Control Variables

We obtain firm-level financial data from Compustat Global measured as of fiscal year-end 2023, ensuring all control variables precede our AI predictions. Our control variables include standard predictors of stock returns and firm value from the asset pricing literature. *FIRMSIZE* is the natural logarithm of total assets. *LEV* is total debt divided by total assets. *ROA* is income before extraordinary items divided by total assets. *LOSS* is an indicator variable equal to one if the firm reported negative net income. *BM* is the book-to-market ratio, calculated as book value of equity divided by market value of equity. *PRE_RETURN* is the stock return during 2023. *RETURN_VOLATILITY* is the standard deviation of daily returns during 2023. *IO* is the percentage of shares held by institutional investors, obtained from the China Stock Market and Accounting Research (CSMAR) database.

For robustness tests and mechanism analyses, we collect additional data. Foreign institutional ownership (*FOREIGN_IO*) comes from CSMAR's detailed ownership files. Actual stock prices on December 31, 2024, used to calculate prediction errors, are obtained from Compustat Global.

2.5 Sample Composition and Descriptive Statistics

Table 1, Panel A provides descriptive statistics for our key variables. The AI prediction variables reveal interesting baseline patterns. *AI_PREDICTION* has a mean of 20.76 RMB with substantial variation (standard deviation = 16.07). The distribution is right-skewed with median (15.33) below mean, and interquartile range spanning 8.32 to 28.46 RMB. Strikingly, *AI_RECOMMEND* has a mean of 0.98, indicating 98% of all AI assessments expected stock price increases. This extreme optimism from both models provides a high baseline against which to measure differential optimism.

Table 1, Panel B, provides more detailed summary statistics on the news coverage for Chinese listed firms in our sample, comparing coverage from Chinese media sources (B.1) and US media sources (B.2). We report four measures: total news articles (*Total News*), articles with negative news (*Negative News*), articles with positive news (*Positive News*), and the difference between negative and positive articles (*Net Negative News*). The bottom row (B.3) reports the difference between US and Chinese source coverage, with significance stars indicating whether the gap is statistically different from zero. On average, U.S. media generate far fewer news articles than Chinese media: the average firm in our sample has 1.68 U.S. articles compared to 26.93 Chinese articles. While both sources produce more negative than positive articles in absolute terms, Chinese media coverage is net negative (2.24 more negative than positive articles per firm), whereas U.S. coverage is slightly net positive (-0.10), meaning marginally more positive than negative articles. All differences between US and Chinese coverage are statistically significant at the 1% level.

These coverage differences result in meaningful information gaps. Returning to Panel A, the mean *NEG_NEWS_GAP* of 0.78 indicates that negative news is heavily concentrated in Chinese media relative to US media, while the mean *NET_NEG_NEWS_GAP* of 0.35 suggests Chinese outlets are, on net, more negative than US outlets. These asymmetries raise the possibility that US-based LLMs like ChatGPT may incorporate less negative news about Chinese companies in their training data than China-based DeepSeek.

Table 1, Panel C reports the industry composition of our sample using the Fama–French 12-industry classification and, within each industry, the average cross-border news gap measures. Manufacturing is the largest sector (24.89%), followed by Business Equipment (20.37%), with sizable representation from Chemicals (8.86%) and Healthcare (8.44%). Consumer Nondurables (7.73%) and Consumer Durables (6.77%) also account for meaningful shares of the sample, while Financials represent 4.38%. Telecommunications is the smallest sector (0.52%).[5]

Panel C also shows substantial cross-industry heterogeneity in cross-border information gaps. For most industries, *NEG_NEWS_GAP* is strongly positive (roughly 0.67 to 0.85), indicating that negative coverage is, on average, more concentrated in Chinese outlets than in U.S. outlets. At the

[5]Panel C counts firm–AI observations (9,956 total). Because each underlying firm appears twice (once per model), the industry shares are identical to those computed at the firm level.

same time, *NET_NEG_NEWS_GAP* varies across industries: Financials stand out with a notably lower *NEG_NEWS_GAP* (0.2618) and *NET_NEG_NEWS_GAP* (0.0896), indicating relatively more balanced cross-border coverage for this sector. This industry-level variation in the information environment motivates our use of industry fixed effects and industry-clustered inference in the ensuing empirical analyses.

3 Empirical Results

3.1 Foreign Bias in AI Predictions

We begin by establishing our central empirical finding: when evaluating Chinese firms, US-based ChatGPT is systematically more optimistic than China-based DeepSeek. This pattern—which we term "foreign bias"—reverses the traditional home bias observed among human investors.

To test for systematic differences in AI predictions, we estimate the following specification:

$$Y_{i,m} = \alpha + \beta \cdot \mathit{US\_AI}_m + \gamma' \mathbf{X}_i + \delta_j + \varepsilon_{i,m} \tag{3}$$

where $Y_{i,m}$ denotes the prediction for firm i by model m. In our main tests, $Y_{i,m}$ is either *AI_PREDICTION*, the predicted stock price for December 31, 2024 (approximately six months after the knowledge cutoff date), or *AI_RECOMMEND*, an indicator equal to one for expected increase in stock price and zero otherwise. The key independent variable, $\mathit{US\_AI}_m$, is an indicator equal to one for predictions generated by ChatGPT 4.1 and zero for those from DeepSeek R1. Thus, the coefficient β captures the systematic difference in predictions between the US-based and China-based models—a positive β indicates that ChatGPT is relatively more optimistic.

The vector $\mathbf{X}_i$ includes firm-level controls measured as of year-end 2023: *FIRMSIZE* (natural logarithm of total assets), *LEV* (total debt divided by total assets), *ROA* (income before extraordinary items divided by total assets), *LOSS* (indicator for negative net income), *BM* (book-to-market ratio), *PRE_RETURN* (stock return during 2023), *RETURN_VOLATILITY* (standard deviation of daily returns during 2023), and *IO* (percentage of shares held by institutional investors). We include industry fixed effects δ_j based on the Fama-French 12-industry classification and cluster standard errors at the industry level.

Table 2 presents the results. Column (1) reveals that ChatGPT predicts systematically higher stock prices than DeepSeek. The coefficient on US_AI is 2.4153 ($p < 0.01$), indicating that ChatGPT's price predictions exceed DeepSeek's by approximately 2.4 RMB per share. This difference is economically substantial, representing 12.53% of the mean actual stock price (19.28 RMB) and approximately 19.68% of the median actual stock price (12.27 RMB). To contextualize this magnitude, a 2.4 RMB difference for the median firm translates to a market capitalization difference of approximately 240 million RMB, assuming 100 million shares outstanding.

The control variables exhibit intuitive relationships with price predictions. More profitable firms (*ROA*) receive higher predictions, while firms with higher book-to-market ratios (*BM*) or prior returns (PRE_RETURN) receive lower predictions, consistent with AI models incorporating growth characteristics and mean reversion into their forecasts.

Column (2) examines directional forecasts, where $AI\_RECOMMEND$ equals one for expected increase in stock price and zero otherwise. The coefficient on US_AI is 0.0135 ($p < 0.01$), indicating ChatGPT issues 1.35 percentage points more expected stock price increase than DeepSeek. While this difference appears modest in absolute terms, it is substantial given the high baseline rate. With DeepSeek recommending buys for 97.55% of firms, ChatGPT's 98.85% buy rate represents a 53% reduction in positive directional forecasts (from 2.45% to 1.15%). Together, the price predictions and directional forecasts indicate substantial relative optimism by the US-based model toward Chinese firms.

3.2 Prediction Accuracy

To ascertain whether ChatGPT's relatively higher forecasts reflect systematic error or superior information, we compare the LLMs' predictions against realized outcomes. If ChatGPT's higher predictions stem from superior information, we would expect them to result in lower forecast errors; if they reflect optimism bias, we would expect greater forecast errors.

We estimate Eq., (3) using measures of prediction accuracy as the dependent variables. We employ two measures: *AI_PREDICTION_ERROR*, defined as the absolute difference between the predicted and actual stock price on December 31, 2024, and *AI_RECOMMEND_ERROR*, defined as the directional classification error $|\mathbf{1}[\text{Buy}] - \mathbf{1}[\text{Price Increase}]|$. In both specifications, a positive coefficient β indicates that ChatGPT exhibits larger prediction errors than DeepSeek.

Table 3 presents the results. In Column (1), the coefficient on *US_AI* is 1.0863 ($p < 0.01$), showing that ChatGPT's price predictions exhibit significantly larger absolute errors than DeepSeek's—approximately 13.3% larger relative to DeepSeek's baseline error. In Column (2), the coefficient on *US_AI* is 0.0074 ($p < 0.10$), indicating higher classification errors in directional forecasts.

Taken together, Table 2 and Table 3 establish the paper's two main stylized facts: ChatGPT exhibits systematic optimism relative to DeepSeek when evaluating Chinese firms, and this optimism translates into lower predictive accuracy. The foreign bias therefore reflects genuine bias rather than superior information processing by the US-based model. We now turn to understanding what drives this pattern.

3.3 Conceptual Framework and Empirical Strategy

What explains ChatGPT's foreign bias? We conceptualize an LLM as implementing a conditional distribution $p_{\theta_{\mathcal{I}}}(\cdot \mid x)$ over outputs given a prompt x, where $\theta_{\mathcal{I}}$ denotes parameters learned from a training information set $\mathcal{I}$. Under this view, cross-model differences in predictions for the same firm can arise from two distinct sources: (i) differences in the information available during training ($\mathcal{I}$), and (ii) differences in how that information is represented in the learned parameters ($\theta_{\mathcal{I}}$) and elicited at inference time. Under this conceptual framing, differences in LLMs' assessments of the same firms could be due to *information-centric* explanations, which emphasize differences in training data availability, and *function-centric* explanations, which emphasize differences in how models process and respond to information.

3.3.1 Information-Centric Explanations

Information-centric explanations posit that prediction differences arise primarily from asymmetries in what information is available to each model during training. Under this view, relevant information about Chinese firms may be sparse or absent from ChatGPT's training corpus relative to DeepSeek's, leading to systematically different assessments.

Several features of the information environment support this possibility. First, Chinese-language news and financial media represent a vast corpus of firm-specific information that may be underrepresented in English-centric training data. Second, US media coverage of Chinese firms is often limited to a subset of larger or more internationally prominent companies, leaving smaller firms with sparse English-language coverage. Third, even when US media do cover Chinese firms, the coverage may emphasize different aspects or carry different sentiment than local Chinese coverage.

Critically, if Chinese-language news contains disproportionately more negative information about Chinese firms than US-language coverage—perhaps because local journalists have better access to firm-specific developments or because US coverage focuses on growth narratives—then a model trained primarily on English sources could systematically underweight negative signals. This may manifest as relative optimism in predictions.

The information-availability mechanism yields several testable predictions. First, ChatGPT's relative optimism should be most pronounced for firms where cross-market information gaps are largest—specifically, firms where Chinese media report more negative news than US media. If ChatGPT's training data draws more heavily on US sources, it would systematically miss unfavorable signals that DeepSeek incorporates for such firms. Second, the bias should attenuate for firms where cross-market information asymmetries are naturally smaller, such as cross-listed firms that receive greater English-language disclosure and analyst attention. Third, providing both models with Chinese news at inference time should substantially reduce or eliminate the prediction gap.

Because prompting can add information to the models' context windows even though it cannot alter learned parameters, equalizing information at inference helps us isolate the role of information set differences. Fourth, because English-language information about American public firms are likely to be more comparably accessible, we expect the LLMs' prediction differences to be small or absent when they analyze US firms.

3.3.2 Function-Centric Explanations

Function-centric explanations focus on differences in how models represent information or respond to prompts, even when similar information is nominally available in training data. We distinguish two sub-channels within this class, corresponding to training-time and inference-time mechanisms.

Under a *parameter encoding* (a training-time) mechanism, Chinese-language information may be present in ChatGPT's training data but weakly encoded in the model's learned parameters. This could occur if Chinese sources receive lower weight during training, if the model's architecture processes non-English text less effectively, or if information from Chinese sources is not as deeply integrated into the model's representations. Under this mechanism, injected news would compete with existing priors encoded in the parameters, potentially only partially attenuating the bias.

Under an *elicitation* mechanism, the two models may respond differently to prompts in ways unrelated to their underlying information or knowledge. For example, ChatGPT might be more prone to optimistic framing in English, might interpret financial prediction prompts differently, or might have been fine-tuned in ways that affect the sentiment of its outputs. Under this mechanism, the bias would vary with prompt language, format, or model variant, but would not necessarily respond to information provision.

3.3.3 Empirical Strategy

The information-centric and function-centric explanations can produce observationally equivalent outcomes in many settings, making it challenging to distinguish between them. Doing so, however, has important implications for understanding LLM behavior more broadly. Because we cannot directly observe the contents of training corpora or inspect how information is encoded in model parameters, LLMs remain, in important respects, black boxes. Nevertheless, we develop empirical strategies that can provide indirect evidence on the likely sources of the foreign bias.

Our empirical strategy proceeds in two stages. First, we conduct a series of tests exploiting cross-sectional and experimental variation in the information environment. These tests are primarily designed to test the likelihood of information-centric explanations, and to distinguish information availability from parameter encoding as a driver of the foreign bias. For example, if providing information at inference time fully eliminates the bias, this suggests the information was missing rather than present but underweighted.

However, these tests do not rule out elicitation differences, which could produce similar patterns if the models respond differently to prompts regardless of their information content. Thus, our second set of tests are designed to address the possibility that elicitation differences between models could explain ChatGPT's foreign bias. We conduct placebo tests using synthetic firm names (where neither model has firm-specific information), vary prompt language, enable real-time web search, test alternative model versions, and examine different forecast horizons. These tests help narrow the scope for function-centric explanations and strengthen the case for information availability as the likely driver.

3.4 Evidence for Information-Centric Mechanisms

We now examine whether the foreign bias is driven by information asymmetries as our conceptual framework predicts. We conduct four tests, exploiting both cross-sectional variation across firms and experimental manipulation of the information environment.

3.4.1 Cross-Border Media Coverage Gaps

If ChatGPT's optimism stems from limited access to Chinese-language news, particularly negative news, its relative optimism should be most pronounced where Chinese media report more negative news than US media. To test this prediction, we augment our baseline specification to include interactions between *US_AI* and measures of cross-border negative news coverage gaps:

$$\begin{aligned} Y_{i,m} &= \alpha + \beta_1 \cdot US\_AI_m \times NEWS\_GAP\_Measure_i + \beta_2 \cdot NEWS\_GAP\_Measure_i + \beta_3 \cdot US\_AI_m \\ &\quad + \gamma' \mathbf{X}_i + \delta_j + \varepsilon_{i,m} \end{aligned} \tag{4}$$

where $Y_{i,m}$ represents our AI price prediction and directional forecast dependent variables, and $NEWS\_GAP\_Measure_i$ captures the extent to which negative news coverage is concentrated in Chinese versus US media for firm i. We employ two measures. *NEG_NEWS_GAP* is defined as the difference between negative news articles from Chinese sources and US sources, scaled by their sum—values near +1 indicate that negative coverage is concentrated in Chinese outlets, while values near −1 indicate concentration in US outlets. *NET_NEG_NEWS_GAP* captures the net negativity gap, defined as the difference between net negative coverage (negative minus positive articles) in Chinese versus US media, scaled by total coverage. Higher values indicate that Chinese outlets are, on net, more negative than US outlets. We report cluster-robust standard errors, clustering by industry to account for potential within-industry correlation in model price and directional forecasts. In untabulated robustness checks, our inferences are unchanged when instead clustering at the firm level or when using heteroskedasticity-robust (White, 1980) standard errors.

Under the information-availability hypothesis, the coefficient β_1 should be positive: ChatGPT's relative optimism should increase when Chinese media report more negative news than US media. This pattern would be consistent with ChatGPT's training data drawing more heavily on US sources, which would underrepresent negative signals for such firms.

Table 4 presents the results. Columns (1)–(2) present results for AI price predictions. In

Column (1), the interaction between *US_AI* and *NEG_NEWS_GAP* is strongly positive (2.7121, $p < 0.01$). This implies that when US outlets report relatively fewer negative stories than Chinese outlets, ChatGPT becomes more optimistic relative to DeepSeek. In Column (2), results are even more pronounced: the interaction between *US_AI* and *NET_NEG_NEWS_GAP* is large (5.5227, $p < 0.01$), confirming that US–China disparities in negative news coverage amplify ChatGPT's upward bias in stock price forecasts.

Columns (3)–(4) examine AI directional forecasts. The interaction coefficients again show positive and significant effects. In Column (3), the interaction of *US_AI* with *NEG_NEWS_GAP* is 0.0273 ($p < 0.10$), and in Column (4), the interaction of *US_AI* with *NET_NEG_NEWS_GAP* is 0.0372 ($p < 0.10$), suggesting that ChatGPT is more likely than DeepSeek to expect higher future prices when US outlets report less negative Chinese news.

Importantly, when we control for these firm-level news gaps, the baseline foreign-bias coefficient becomes smaller and, in all specifications, statistically insignificant, consistent with coverage gaps mediating the cross-model prediction differences. Similarly, the main effects of *NEG_NEWS_GAP* and *NET_NEG_NEWS_GAP* are statistically insignificant, indicating that cross-border coverage gaps do not systematically affect DeepSeek's predictions. This pattern is consistent with our information-availability hypothesis: DeepSeek has access to Chinese news regardless of US coverage levels, so the cross-border gap is irrelevant for its predictions. The gap only matters for ChatGPT, which may lack Chinese-language sources, and ChatGPT's relative optimism emerges precisely when Chinese firms receive disproportionately negative coverage in Chinese outlets compared to US outlets.

3.4.2 Cross-Listed Firms

Cross-listed firms provide another natural test of the information-asymmetry mechanism. Chinese firms listed on US exchanges are subject to broader disclosures, bilingual filings, and more extensive analyst and media coverage across markets, naturally narrowing the information gap between US and Chinese sources. If cross-border information frictions drive the foreign bias, the effect should weaken for these firms. We test this by estimating:

$$Y_{i,m} = \alpha + \beta_1 \cdot US\_AI_m + \beta_2 \cdot US\_AI_m \times CROSS\_LISTED_i + \beta_3 \cdot CROSS\_LISTED_i + \gamma' \mathbf{X}_i + \delta_j + \varepsilon_{i,m} \quad (5)$$

where $CROSS\_LISTED_i$ is an indicator equal to one if firm i is cross-listed on a US exchange. A negative coefficient β_2 would indicate that ChatGPT's relative optimism is attenuated for cross-listed firms, consistent with the information-availability mechanism.

Table 5 presents the results. Consistent with this prediction, the interaction term $US\_AI \times CROSS\_LISTED$ is negative and statistically significant for both dependent variables. The coefficient is -0.7660 ($p < 0.01$) for price predictions and -0.0082 ($p < 0.01$) for directional forecasts. Meanwhile, the main effect of US_AI remains positive (2.4467 for prices and 0.0139 for directional forecasts), indicating that ChatGPT's optimism persists but is significantly attenuated—approximately 30% smaller for price predictions and 60% smaller for directional forecasts—for cross-listed Chinese firms. Finally, the positive coefficient on $CROSS\_LISTED$ (0.5600, $p < 0.01$) indicates that cross-listed firms receive higher price predictions overall, consistent with their greater transparency and international visibility.

The attenuation of the foreign bias for cross-listed Chinese firms provides further support that the bias arises from differences in information environments rather than inherent model-specific optimism. When informational asymmetries between the US and Chinese markets are naturally smaller, so too is the magnitude of ChatGPT's foreign bias.

3.4.3 Closing the News Gap

Our most direct test of the information-availability mechanism involves experimentally manipulating the information available to both models at inference time. Table 6 reports results of an intervention in which we augment the prompt for both ChatGPT and DeepSeek with Chinese-language news about each firm drawn from CFND over June 2022 to June 2024. Because providing the full text of all articles exceeds practical limits, we instead supply the *abstracts* (i.e., the short summary fields) for *all* available Chinese news articles in that window.[6] This design directly targets the hypothesized information gap: it equalizes access to local Chinese news at prediction time while leaving the models' learned parameters unchanged.

Panel A, Table 6, shows that once both models are conditioned on the same Chinese news abstracts, the baseline foreign bias completely disappears. The coefficient on *US_AI* becomes statistically indistinguishable from zero (at the 10% level) for both price predictions (-0.7980) and directional forecasts (-0.0022), and economically insignificant compared to the baseline effects of 2.4153 and 0.0135, respectively, from Table 2. Once we ensure that both ChatGPT and DeepSeek have access to Chinese news in the inference context, ChatGPT's bias is essentially eliminated.

To clarify which model drives this convergence, Panel B reports an analysis that compares the baseline LLM price and directional forecasts with their news-augmented counterparts. We stack the two sets of predictions and estimate an interaction specification, using an indicator (*NEWS_INJECTED*) for the predictions based on news-augmented prompts and an interaction between *NEWS_INJECTED* and *US_AI*. This specification is similar to a difference-in-differences specification in which *NEWS_INJECTED* is the treatment and *US_AI* is the affected group.

In Panel B, Table 6, the positive and significant *US_AI* coefficient reproduces the foreign-bias estimate of Table 2: in the absence of or with news-augmentation intervention, ChatGPT shows relative optimism for Chinese firms. In contrast, the main effect of *NEWS_INJECTED* is zero, which implies that augmenting the prompt with Chinese news abstracts has no detectable average effect on DeepSeek's output. Notably, the interaction term $US\_AI \times NEWS\_INJECTED$ is strongly negative and highly significant: -2.410 (s.e. 0.046) for price predictions and -0.009 (s.e. 0.002) for directional forecasts.

Together, the *NEWS_INJECTED* main effect and the interaction effect suggests that news

[6] In an alternative implementation, we instead randomly sample five full-text Chinese news articles from the same June 2022–June 2024 window and provide the full text of those articles in the prompt. Untabulated results are very similar, indicating that our findings are not driven by using abstracts rather than full articles.

injection primarily moves ChatGPT: it revises downward by approximately 3.21 RMB in price predictions and 1.57 percentage points in the directional forecast outcome, offsetting the baseline optimism gap of 2.42 RMB and 1.35 percentage points.

The fact that the baseline foreign-bias collapses under news augmentation is helpful in distinguishing between information availability and parameter encoding explanations. Prompting fundamentally cannot alter a model's pre-trained weights—it can only provide new information to process. If the bias reflected Chinese information being present but systematically underweighted in ChatGPT's parameters, one might expect injected news to compete with existing priors, only partially attenuating the foreign bias. Instead, we find that information provision alone eliminates it. Although we cannot directly observe how information is encoded or retrieved within the model, the fact that the prediction changes resulting from news-augmentation are concentrated in ChatGPT rather than DeepSeek further supports the information-availability interpretation.

3.4.4 US Firms: A Symmetric Information Benchmark

The information-availability mechanism predicts a foreign-bias asymmetry: such a bias should emerge when evaluating firms from the information-disadvantaged market but not when evaluating firms where both models have comparably rich information. For US firms, English-language coverage is abundant and accessible to both models. Moreover, as noted in footnote 3, allegations that DeepSeek was trained in part on ChatGPT outputs suggest that, if true, information ChatGPT possesses about US firms may also be effectively embedded in DeepSeek's training data. Both channels point toward more symmetric information environments when analyzing US firms.

Table 7 tests this prediction by repeating the baseline analysis for a sample of 5,844 US firms. After collecting the predictions about these US firms from ChatGPT and DeepSeek, we estimate the baseline regression (Eq., (3)). The coefficients on US_AI are small and statistically insignificant in both columns: -0.1539 for price predictions and -0.0472 for directional forecasts. These null results stand in stark contrast to the statistically strong and positive coefficients observed for Chinese firms, confirming that the foreign bias is specific to cross-border predictions where information asymmetries are severe.

3.5 Assessing Elicitation-Based Explanations

While the evidence presented thus far favors the information-availability mechanism, another potential explanation behind the observed LLM foreign bias could be elicitation differences between models. Under this alternative interpretation, the models might respond differently to our prompts in ways that generate systematic prediction differences unrelated to their underlying information or knowledge.

For example, ChatGPT might be more prone to optimistic framing when responding to English prompts, might interpret financial prediction tasks differently, or might have been fine-tuned in ways that affect the sentiment of its outputs. If such elicitation differences were driving our results, we would expect the bias to vary with prompt language or format, to appear even when both models lack firm-specific information, or to attenuate with model updates that improve reasoning capabilities and become less reliant on prompting cues.

We conduct several tests designed to assess the scope for elicitation-based explanations.

3.5.1 **Synthetic Firms Placebo**

We begin with an experiment designed to test whether the models respond differently to Chinese-sounding company names in the prompt, holding information differences constant. The logic is as follows: if ChatGPT's relative optimism is triggered by elicitation cues, such as recognizing a firm as Chinese and responding with systematically different sentiment, then the bias should emerge even when neither model has any real information about the firm. If, instead, the bias requires genuine information asymmetries, it should disappear when both models are equally uninformed.

Table 8 implements this test using a placebo sample of 100 synthetic Chinese firm names. These are fictitious firms with Chinese-sounding names but no corresponding real-world entities or media coverage. By design, there is no underlying firm-specific information for either model to differentially access; any systematic bias in this setting would therefore be consistent with name-triggered elicitation rather than information availability.

After collecting the predictions about these synthetic firms from ChatGPT and DeepSeek, we again estimate the baseline regression (Eq., (3)). Table 8 shows that the coefficient on *US_AI* is small and statistically insignificant (at the 10% level) for both price predictions and directional forecasts. This null result is inconsistent with a pure elicitation explanation, which would predict that Chinese-sounding names alone might trigger differential responses between models.

Moreover, the results of this test further support the information availability hypothesis. Together, our empirical results thus far suggest ChatGPT's foreign bias requires genuine cross-border information asymmetry. When firm-specific information is absent and both models operate under symmetric informational conditions, the excess optimism disappears.

3.5.2 Robustness Tests

While the synthetic firms placebo rules out name-triggered elicitation as a driver of the foreign bias, other elicitation-based mechanisms remain possible. ChatGPT and DeepSeek may differ in how they respond to English versus Mandarin prompts, how they interpret financial prediction tasks, or how their fine-tuning affects output sentiment. We conduct four additional tests to assess these possibilities, reported in Table 9. Each of these tests below implements our baseline regression analysis (Eq., (3)), but applies different prompts or underlying models to the same 4,978 Chinese firms as in Table 2.

We first examine whether prompt language drives the results. If ChatGPT exhibits greater optimism because it responds differently to English-language prompts—perhaps due to language-specific fine-tuning—then delivering prompts in Mandarin should attenuate or eliminate the bias. Panel A shows that the foreign bias persists when prompts are delivered in Mandarin: the coefficient on *US_AI* remains positive and significant for both price predictions (1.5173, $p < 0.05$) and directional forecasts (0.0135, $p < 0.10$). The magnitude of the price-prediction effect is attenuated relative to the baseline in Table 2, by about 37%; on the other hand, the magnitude of the directional forecast effect remains the same. Yet, the vast majority of the foreign bias clearly survives the switch to Mandarin prompts, suggesting it is not simply an artifact of prompt language.

We next examine whether inference-time scaffolding—either through expanded information access or enhanced reasoning capabilities—changes the magnitude of the bias. Panel B reports results when online search is enabled for both models, which expands the information available to the LLM at prediction time. If the foreign bias is purely elicitation-driven—reflecting how the models respond to prompts rather than what information they possess—then expanding information access may not systematically attenuate the bias. In contrast, if the bias reflects missing information, then allowing models to search the web should reduce the gap.

Table 9, Panel B shows that enabling online search does attenuate, but not eliminate, the bias for price predictions: the coefficient on US_AI in column (1) falls to 0.5338 ($p < 0.05$), or a 78% reduction of the baseline bias. On the other hand, the directional forecast effect persists. We interpret these results to be more consistent with an information-availability explanation than a pure elicitation story. In addition, the persistence of a foreign bias effect suggests that ad hoc retrieval does not fully close the models' effective information gap.

We next test whether the bias attenuates with improved model capabilities. Recent advances in LLM development have produced reasoning-oriented model variants that are designed to improve multi-step reasoning performance. To the extent such variants reduce sensitivity to superficial prompting cues, a purely elicitation-driven mechanism could suggest the elimination of the AI foreign-bias when prompting such reasoning-oriented models.

Table 9, Panel C reports results using ChatGPT-5 nano, a smaller reasoning-oriented ChatGPT variant. We use the nano version for processing efficiency considerations. Because ChatGPT-5 does not allow setting the temperature parameter to zero (unlike DeepSeek R1), we take the average of 50 prompts for price predictions and the median for directional forecasts (effectively implementing a majority-vote classification).

Panel C shows that the coefficient on US_AI remains positive and significant: 1.6622 ($p < 0.05$) for price predictions and 0.0221 ($p < 0.01$) for directional forecasts. The magnitude of the price-prediction effect is attenuated relative to the baseline in Table 2, by about 31%.[7] On the other hand, the magnitude of the directional forecast effect actually increases. Similar to our analysis in Panel A, the vast majority of the foreign bias survives the switch to a reasoning-based US model. To the extent reasoning-oriented models reduce reliance on superficial prompting cues, the persistence of the bias in such a model suggests that the foreign bias effect is unlikely to be driven by elicitation differences.

Finally, we test whether the bias is confined to short-horizon predictions. If elicitation differences cause ChatGPT to respond more optimistically to near-term forecasting tasks, extending the prediction horizon should attenuate the effect. Panel D extends the forecast horizon to 12 months. That is, we ask both models to make predictions for prices and directional forecasts with a target end date of June 30, 2025. The coefficient on US_AI remains positive and significant: 1.8490 ($p < 0.05$) for price predictions and 0.0207 ($p < 0.01$) for directional forecasts. The persistence of the bias at longer horizons indicates it is not an artifact of short-term prediction framing.

The tests in Table 8 and Table 9 substantially narrow the scope for elicitation-based explanations. The foreign-bias pattern is stable across prompt language, synthetic firm placebos, model variants, retrieval access, and forecast horizons. These results strengthen the interpretation that information availability, rather than differences in how models respond to prompts, as a more likely driver of the documented foreign bias.

3.6 Discussion

A fundamental challenge in interpreting our results is that LLMs remain, in important respects, black boxes. We cannot directly observe the contents of ChatGPT's or DeepSeek's training corpora,

[7] In untabulated results, we find that taking the median of the 50 price predictions yields very similar results. The US_AI remains positive (1.4373) and significant at the 5% level.

the weighting schemes applied during training, or the specific objectives used in post-training fine-tuning. Any conclusion about why the foreign bias arises must therefore be inferred indirectly from observable patterns in model outputs.

With this caveat in mind, we summarize the weight of evidence. The foreign bias: (i) intensifies when Chinese media report more negative news than US media; (ii) attenuates for cross-listed firms, which have greater salience in the US and coverage by the US media; (iii) disappears entirely when both models receive Chinese news at inference time; (iv) is absent when analyzing US firms where information environments are likely to be more symmetric; (v) is absent for synthetic firms where neither model possesses real information; and (vi) persists across prompt language, reasoning-oriented model variants, and forecast horizons.

The first four patterns are consistent with information-centric explanations; the fifth rules out name-triggered elicitation; and the sixth narrows the scope for other elicitation-based mechanisms. On balance, the evidence all points to *effective* information availability/access as a first-order driver of the AI foreign-bias we document. While we cannot rule out that some residual bias reflects parameter encoding or elicitation differences, such mechanisms appear insufficient to explain the core phenomenon.

4 Conclusion

This paper documents a novel *foreign bias* in AI-generated financial predictions: when evaluating Chinese listed firms, ChatGPT is systematically more optimistic than a China-developed frontier model (DeepSeek) while being less accurate ex post. This pattern reverses the traditional home bias observed in human investors, suggesting that LLMs' biases need not mirror human psychology, even when trained on human-generated text.

Our evidence is most consistent with foreign bias arising from *spatial* information-set distortions. ChatGPT's excess optimism is concentrated precisely where cross-market information gaps are plausibly largest (firms with limited US coverage), attenuated where those gaps are smaller (cross-listed firms with greater English-language disclosure), and absent in settings where both models have comparably rich information (US firms). In addition, supplying firm-specific Chinese news at inference time eliminates ChatGPT's excess optimism. Because prompting can add information to the model's context but cannot alter learned parameters, this intervention strongly suggests that differences in effective information availability drive the foreign-bias.

A natural question is why such biases may persist even for state-of-the-art systems. Expanding, balancing, and auditing global coverage can be costly and operationally difficult: high-quality financial information is fragmented across languages, paywalls, licensing regimes, and uneven disclosure standards, and continuous retraining or curation to close coverage gaps may compete with other engineering priorities. Consistent with our finding that enabling web browsing only partially attenuates the bias, ad hoc search may not reliably recover the most relevant local-language information or integrate it into forecasts without careful prompting. Together, these constraints imply that information-set distortions may be persistent features of frontier LLMs.

More broadly, our findings speak to AI development in an increasingly multi-polar landscape. As different jurisdictions build frontier models shaped by different informational environments, the same firm may receive systematically different assessments depending on which model is used. Rather than acting as neutral information technology, LLMs can introduce *model-specific* perspectives that interact with existing cross-border information frictions. This raises a practical concern: the diffusion of AI tools may not automatically democratize sophisticated analysis globally but could instead create new disparities based on which models and which information sources users can access and incorporate.

For practitioners, the implication is straightforward. LLM outputs in cross-border valuation and prediction tasks should be treated as potentially *systematically* skewed. Mitigating these distortions may require explicit local-information supplementation (e.g., by including local-language sources) and validation against alternative models or benchmarks. For policymakers and platform designers, our results highlight the importance of transparency around training data provenance, language coverage, and retrieval protocols, as well as the value of standardized audits that stress-test models across jurisdictions and information environments.

In sum, the foreign bias we document illustrates how exogenous and often unobserved differences in informational environments can generate unexpected and economically meaningful distortions in LLM predictions. As AI tools increasingly mediate cross-border investment decisions, measuring, monitoring, and mitigating information-set distortions will be essential to maintain market efficiency and protect investors.

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Exhibit A. Prompts Used in the Analysis

This exhibit reports the exact prompts (temperature = 0) used to elicit stock price predictions and directional forecasts from large language models across the main analysis and robustness tests. Note: In the actual API calls, placeholders such as {`name`} and {`ticker`} were replaced with the specific company name and ticker symbol.

Panel A. Prompt Used for the Main Analysis (Table 2)

```
Here is a firm called {name} (ticker: {ticker}), listed in China. As a professional
financial analyst:

  1. Predict the firm's stock price on December 31, 2024 (three decimal places).
  2. Predict whether the stock price will increase from the end of June to December
     31, 2024 (1 = increase, 0 = not increase).

Do not use current online data or random numbers. Output only a JSON object, without
any additional text or explanation.
```

Panel B. Prompt Used for Closing the News Coverage Gap (Table 6)

```
Here is a firm called {name} (ticker: {ticker}), listed as an A-share in China.
Below is a collection of Chinese-language news articles related to this firm
published between June 2022 and June 2024.
[Chinese news abstracts inserted here]
Please act as a professional financial analyst and evaluate this firm based on the
information above:

  1. Predict the firm's stock price on December 31, 2024 (three decimal places).
  2. Predict whether the stock price will increase from the end of June to December
     31, 2024 (1 = increase, 0 = not increase).

Do not use current online data or random numbers. Output only a JSON object, without
any additional text or explanation.
```

Panel C. Prompt Used for U.S. Firms (Table 7)

```
Here is a firm called {name} (ticker: {ticker}), listed in the United States. Please act as a professional financial analyst and evaluate this firm based on the information above:

1. Predict the firm's stock price on December 31, 2024 (three decimal places).
2. Predict whether the stock price will increase from the end of June to December 31, 2024 (1 = increase, 0 = not increase).

Do not use current online data or random numbers. Output only a JSON object, without any additional text or explanation.
```

Panel D. Prompt Used for Mandarin-Language Elicitation Test (Table 9, Panel A)

```
以下是一家名为{name}（股票代码：{ticker}）的公司，在中国A 股市场上市。请你作为一名专业的金融分析师，基于你已有的知识对该公司进行评估：

1. 预测该公司在2024 年12 月31 日的股票价格（保留三位小数）。
2. 判断该公司股票价格是否会从2024 年6 月底上涨至2024 年12 月31 日（1 = 上涨，0 = 不上涨）。

请勿使用当前的在线数据或编造数据。仅输出一个JSON 对象，不要包含任何额外的文字或解释。
```

Panel E. Prompt Used for Alternative Prediction-Horizon Test (Table 9, Panel D)

```
Here is a firm called {name} (ticker: {ticker}), listed as an A-share in China. Please act as a professional financial analyst and evaluate this firm based on your existing knowledge:

1. Predict the firm's stock price on June 30, 2025 (three decimal places).
2. Indicate whether the stock price will increase from the end of June 2024 to June 30, 2025 (1 = increase, 0 = not increase).

Do not use current online data or perform any external search. Return only a JSON object, without any additional text or explanation.
```

Appendix A Description of Variables

This table defines variables used in our analyses. Variable definitions are based on data obtained from ChatGPT 4.1, DeepSeek R1, RavenPack News Analytics, CFND, Compustat, CSMAR, and Google, as described in Section 2.

Variable	Description	Source
	Key Variables of Interest	
AI_PREDICTION	Stock price prediction of gen AI for a Chinese firm for Dec 31, 2024.	ChatGPT 4.1, DeepSeek R1
AI_RECOMMEND	Indicator = 1 if gen AI forecasts a stock price increase in the June 30–Dec 31, 2024 period; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
AI_PREDICTION_ERROR	|AI predicted price – actual price at Dec 31, 2024|.	ChatGPT 4.1, DeepSeek R1, Compustat
AI_RECOMMEND_ERROR	$\lvert \mathbf{1}[\text{Buy}] - \mathbf{1}[\text{Price Increase}] \rvert$, where $\mathbf{1}[\text{Price Increase}]$ is an indicator of whether the stock price increased from June 30 to Dec 31, 2024.	ChatGPT 4.1, DeepSeek R1, Compustat
NEG_NEWS_GAP	the number of negative news from Chinese media minus the number of negative news from US media divided by their total number during the three-years period prior to the end of training period: i.e., 2022.06.30 to 2024.06.30. It is set to be zero if the value is missing.	RavenPack News Analytics, CFND
NET_NEG_NEWS_GAP	the number of negative news minus the number of positive news from Chinese media minus the number of negative news minus the number of positive news from US media, divided by their total number, during the three-years period prior to the end of training period: i.e., 2022.06.30 to 2024.06.30.	RavenPack News Analytics, CFND
US_AI	Indicator = 1 if the gen AI is the US AI; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
	Firm-Level Control Variables	
BM	Book-to-market ratio in year $t-1$.	Compustat
FIRMSIZE	Log of total assets in year $t-1$.	Compustat
IO	Percentage of institutional ownership in year $t-1$.	CSMAR
LEV	Year-end ratio of total liabilities to total assets in year $t-1$.	Compustat
LOSS	Indicator = 1 if net income is negative in year $t-1$; 0 otherwise.	Compustat
PRE_RETURN	Annual stock returns in year $t-1$.	Compustat
RETURN_VOLATILITY	Std. dev. of daily stock returns in year $t-1$.	Compustat
ROA	Income before extraordinary items / total assets in year $t-1$.	Compustat

Variable	Description	Computation
	Variables in Other Tests	
AI_PREDICTION_ALT	AI prediction for Chinese firm for June 30, 2025.	ChatGPT 4.1, DeepSeek R1
AI_RECOMMEND_ALT	Indicator = 1 if AI recommends purchase for June 30, 2024–June 30, 2025, for a Chinese firm; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
AI_PREDICTION_CHATGPT5	AI prediction for Chinese firm for Dec 31, 2024, using the average of 50 random draws from ChatGPT 5 Nano.	ChatGPT 5 Nano, DeepSeek R1
AI_RECOMMEND_CHATGPT5	Indicator = 1 if AI recommends purchase for June 30, 2024–Dec 31, 2024, for a Chinese firm, using the median of 50 random draws from ChatGPT 5 Nano; 0 otherwise.	ChatGPT 5 Nano, DeepSeek R1
AI_PREDICTION_INJECTED	AI prediction for Dec 31, 2024, after feeding Chinese media news if available.	ChatGPT 4.1, DeepSeek R1, CFND
AI_RECOMMEND_INJECTED	Indicator = 1 if AI recommends purchase for June 30–Dec 31, 2024, after feeding Chinese media news; 0 otherwise.	ChatGPT 4.1, DeepSeek R1, CFND
NEWS_INJECTED	Indicator = 1 if the prediction is made after the gen AI is provided with all available Chinese news articles abstract for the firm from June 30, 2022 to June 30, 2024 in the prompt; 0 otherwise.	ChatGPT 4.1, DeepSeek R1, CFND
AI_PREDICTION_US	AI prediction for US firm for Dec 31, 2024.	ChatGPT 4.1, DeepSeek R1
AI_RECOMMEND_US	Indicator = 1 if AI recommends purchase for June 30–Dec 31, 2024, for a US firm; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
AI_PREDICTION_Mandarin	AI prediction for Chinese firm for Dec 31, 2024, *using Mandarin prompt*.	ChatGPT 4.1, DeepSeek R1
AI_RECOMMEND_Mandarin	Indicator = 1 if AI recommends purchase for June 30, 2024–Dec 31, 2024, for a Chinese firm, *using Mandarin prompt*; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
AI_PREDICTION_Web	AI prediction for Chinese firm for June 30, 2025, *using web search*.	ChatGPT 4.1, DeepSeek R1
AI_RECOMMEND_Web	Indicator = 1 if AI recommends purchase for June 30, 2024–June 30, 2025, for a Chinese firm, *using web search*; 0 otherwise.	ChatGPT 4.1, DeepSeek R1
CROSS_LISTED	Indicator = 1 if the Chinese firm is a cross-listed firm; 0 otherwise.	Compustat

Table 1. Summary Statistics

This table reports descriptive statistics for the main variables in our sample of Chinese firms used in the baseline analysis. Panel A provides the distribution of the primary dependent and independent variables across 9,956 firm–AI observations, including AI price predictions, directional forecasts, and key firm characteristics. Panel B reports means and standard errors of news coverage for Chinese listed firms in the sample, comparing coverage from Chinese media sources (B.1) and US media sources (B.2). We report four measures: total news articles (*Total News*), articles with negative news (*Negative News*), articles with positive news (*Positive News*), and the difference between negative and positive articles (*Net Negative News*). Panel B.3 reports the difference between US and Chinese source coverage, with significance stars indicating whether the gap is statistically different from zero. Panel C reports the industry distribution of firms in our sample by Fama-French 12 industry classification and the average news gap measures. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles.

Panel A: Distribution of Main Variables

Variable	*N*	*Mean*	*SD*	*p10*	*p25*	*p50*	*p75*	*p90*
AI_PREDICTION	9,956	20.76	16.07	5.33	8.32	15.33	28.46	45.33
AI_RECOMMEND	9,956	0.98	0.13	1.00	1.00	1.00	1.00	1.00
US_AI	9,956	0.50	0.50	0.00	0.00	0.50	1.00	1.00
NEG_NEWS_GAP	9,956	0.78	0.45	0.00	1.00	1.00	1.00	1.00
NET_NEG_NEWS_GAP	9,956	0.35	0.59	-0.38	-0.02	0.27	1.00	1.00
FIRM_SIZE	9,956	8.60	1.40	7.13	7.62	8.29	9.30	10.50
ROA	9,956	0.02	0.06	-0.05	0.00	0.03	0.05	0.09
LEVERAGE	9,956	0.42	0.22	0.13	0.24	0.40	0.57	0.71
LOSS	9,956	0.21	0.41	0.00	0.00	0.00	0.00	1.00
BM	9,956	0.57	0.46	0.18	0.29	0.45	0.70	1.02
PRE_RETURN	9,956	0.03	0.32	-0.32	-0.17	0.00	0.17	0.39
RETURN_VOLATILITY	9,956	2.36	0.88	1.36	1.71	2.19	2.88	3.59
IO	9,956	41.11	25.12	7.28	19.63	40.43	61.78	75.79

Panel B: New Coverage by Media Source

	Total News	*Negative News*	*Positive News*	*Net Negative News*
B.1: Chinese Sources				
Mean	26.9339	14.5882	12.3457	2.2425
	(0.6284)	(0.3348)	(0.4082)	(0.4032)
B.2: US Sources				
Mean	1.6826	0.7921	0.8905	-0.0984
	(0.0748)	(0.0415)	(0.0406)	(0.0339)
B.3: Difference (US − Chinese)				
Difference	-25.2513***	-13.7961***	-11.4552***	-2.3409***
	(0.6216)	(0.3334)	(0.4055)	(0.4059)

Panel C: Average News Gap by Fama-French 12-industry classification

Industry	*Number of Firms*	*%*	*NEG_NEWS_GAP*	*NET_NEG_NEWS_GAP*
Consumer Nondurables	770	7.73	0.8506	0.3077
Consumer Durables	674	6.77	0.7820	0.3611
Manufacturing	2,478	24.89	0.7900	0.3628
Energy	124	1.25	0.6762	0.1602
Chemicals	882	8.86	0.8122	0.3473
Business Equipment	2028	20.37	0.8035	0.4459
Telecommunications	52	0.52	0.8385	0.3449
Utilities	176	1.77	0.8353	0.3321
Wholesale and Retail	370	3.72	0.7921	0.3355
Healthcare	840	8.44	0.8294	0.3134
Financials	436	4.38	0.2618	0.0896
Others	1,126	11.31	0.7747	0.3304

Table 2. ChatGPT vs. DeepSeek: Price Predictions and Directional Forecasts

This table presents baseline regressions comparing six-month-ahead stock price predictions (Column 1) and directional forecasts (Column 2) for Chinese firms generated by ChatGPT (US-based) and DeepSeek (China-based). The key independent variable, *US AI*, equals one for ChatGPT predictions and zero for DeepSeek. Control variables include firm size, leverage, profitability, book-to-market ratio, prior returns, return volatility, loss indicator, and institutional ownership, with industry fixed effects. The coefficients on *US AI* capture the magnitude of the documented "foreign bias." All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1) *AI_PREDICTION*	(2) *AI_RECOMMEND*
US_AI	2.4153***	0.0135***
	(0.722)	(0.004)
FIRM_SIZE	0.7579	-0.0025
	(0.788)	(0.002)
ROA	33.7108***	-0.0187
	(7.558)	(0.054)
LEVERAGE	-15.9897***	-0.0235***
	(3.649)	(0.007)
LOSS	0.6487	-0.0114
	(1.013)	(0.012)
BM	-4.1948***	-0.0349***
	(1.216)	(0.009)
PRE_RETURN	-8.6622***	0.0008
	(0.951)	(0.008)
RETURN_VOLATILITY	4.9367***	-0.0009
	(0.778)	(0.002)
IO	0.0702*	0.0001
	(0.033)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.3071	0.0325

Table 3. Prediction Errors

This table compares the accuracy of ChatGPT and DeepSeek predictions against realized outcomes. Column 1 reports absolute price prediction errors, and Column 2 reports directional forecast errors, defined as $|\mathbf{1}[\text{Buy}] - \mathbf{1}[\text{Price Increase}]|$, where $\mathbf{1}[\text{Price Increase}]$ is an indicator of whether the stock price increased from June 30 to Dec 31, 2024. Positive coefficients on *US AI* indicate that ChatGPT is less accurate than DeepSeek, consistent with its optimism reflecting systematic error rather than superior information. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1)	(2)
	AI_PREDICTION_ERRORS	*AI_RECOMMEND_ERRORS*
US_AI	1.0863***	0.0074*
	(0.318)	(0.004)
FIRM_SIZE	0.3676	0.0121
	(0.454)	(0.013)
ROA	24.6574***	0.9947***
	(5.694)	(0.155)
LEVERAGE	-6.2347***	0.0307
	(1.926)	(0.029)
LOSS	1.8352**	0.0532***
	(0.657)	(0.015)
BM	-2.8381**	0.0875**
	(0.946)	(0.030)
PRE_RETURN	-1.3305	0.0206
	(0.815)	(0.035)
RETURN_VOLATILITY	2.6255***	0.0431**
	(0.566)	(0.014)
IO	0.0462*	0.0003
	(0.022)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.1487	0.0565

Table 4. Role of Media Coverage in AI Prediction: Gap in News Coverage

This table examines whether cross-border media coverage asymmetries explain the foreign bias in AI-generated predictions. We regress predicted prices (Columns 1–2) and directional forecasts (Columns 3–4) on the interaction between *US AI* and news gap measures: *NEG_NEWS_GAP* and *NET_NEG_NEWS_GAP*. The news gap variables measure the relative prevalence of U.S. versus Chinese negative news coverage for each firm. Positive interaction coefficients indicate that when negative news coverage is more concentrated in Chinese outlets (and thus relatively scarce in U.S. outlets), ChatGPT's relative optimism increases. All continuous variables are winsorized at the 1st and 99th percentiles. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1)	(2)	(3)	(4)
	AI_PREDICTION		*AI_RECOMMEND*	
US_AI×NEG_NEWS_GAP	2.7121***		0.0273*	
	(0.670)		(0.014)	
US_AI×NET_NEG_NEWS_GAP		5.5227***		0.0372*
		(0.971)		(0.018)
NEG_NEWS_GAP	-0.5303		0.0135	
	(0.632)		(0.010)	
NET_NEG_NEWS_GAP		-0.2494		-0.0012
		(0.391)		(0.004)
US_AI	0.3079	0.4824	-0.0078	0.0004
	(0.225)	(0.343)	(0.012)	(0.007)
FIRM_SIZE	0.7959	1.0361	-0.0013	-0.0006
	(0.797)	(0.801)	(0.003)	(0.003)
ROA	33.1979***	33.6656***	-0.0355	-0.0190
	(7.573)	(7.523)	(0.053)	(0.053)
LEVERAGE	-16.0780***	-15.7076***	-0.0264***	-0.0216***
	(3.703)	(3.650)	(0.006)	(0.006)
LOSS	0.6011	0.5157	-0.0130	-0.0123
	(0.988)	(0.984)	(0.012)	(0.012)
BM	-4.2290***	-4.2135***	-0.0360***	-0.0350***
	(1.239)	(1.184)	(0.009)	(0.009)
PRE_RETURN	-8.6093***	-8.5130***	0.0026	0.0019
	(0.933)	(0.915)	(0.008)	(0.008)
RETURN_VOLATILITY	4.9214***	4.8234***	-0.0014	-0.0017
	(0.780)	(0.762)	(0.001)	(0.002)
IO	0.0690*	0.0686*	0.0001	0.0001
	(0.033)	(0.032)	(0.000)	(0.000)
Industry FE	Yes	Yes	Yes	Yes
Observations	9,956	9,956	9,956	9,956
Adj R^2	0.3089	0.3248	0.0420	0.0442

Table 5. Cross-Listing Firms

This table investigates whether foreign bias persists is less pronounced for cross-listed Chinese firms. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1)	(2)
	AI_PREDICTION	*AI_RECOMMEND*
US_AI	2.4467***	0.0139***
	(0.716)	(0.004)
US_AI×*CROSS_LISTED*	-0.7660***	-0.0082***
	(0.046)	(0.002)
CROSS_LISTED	0.5600***	-0.0013
	(0.046)	(0.002)
FIRM_SIZE	0.7471	-0.0026
	(0.784)	(0.002)
ROA	33.6858***	-0.0068
	(7.636)	(0.053)
LEVERAGE	-15.9575***	-0.0229***
	(3.644)	(0.005)
LOSS	0.6376	-0.0103
	(1.017)	(0.012)
BM	-4.1986***	-0.0334***
	(1.201)	(0.010)
PRE_RETURN	-8.6770***	0.0005
	(0.949)	(0.008)
RETURN_VOLATILITY	4.9266***	-0.0007
	(0.777)	(0.002)
IO	0.0704*	0.0001*
	(0.033)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.3085	0.0425

Table 6. Closing the News Gap

This table reports results from the information intervention in which ChatGPT and DeepSeek were both provided with the same abstracts of all Chinese news articles (from June 2022 to June 2024) about each firm prior to making stock price predictions (column 1) or directional forecasts (column 2). Panel B stacks baseline and news-augmented predictions (*AI_PREDICTION_STACKED*) and estimates an interaction model, akin to a difference-in-differences specification, with an indicator for news augmentation (*NEWS_INJECTED*) and its interaction with *US_AI*. The coefficient on *NEWS_INJECTED* captures the effect of news on DeepSeek; the interaction captures the incremental effect on ChatGPT relative to DeepSeek. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

Panel A: News-Augmented Predictions

	(1) *AI_PREDICTION_INJECTED*	(2) *AI_RECOMMEND_INJECTED*
US_AI	-0.7980	-0.0022
	(0.770)	(0.003)
FIRM_SIZE	0.6187	-0.0047
	(0.808)	(0.003)
ROA	33.5798***	-0.0372
	(8.068)	(0.069)
LEVERAGE	-16.5883***	-0.0251***
	(3.731)	(0.005)
LOSS	0.4735	-0.0132
	(1.037)	(0.015)
BM	-4.0468***	-0.0601***
	(1.211)	(0.014)
PRE_RETURN	-9.0368***	-0.0039
	(0.986)	(0.010)
RETURN_VOLATILITY	5.1877***	-0.0014
	(0.818)	(0.002)
IO	0.0752*	0.0002*
	(0.035)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.3060	0.0492

Table 6 [Continued]

Panel B: Comparing Baseline and News-Augmented Predictions

	(1)	(2)
	AI_PREDICTION_STACKED	*AI_RECOMMEND_STACKED*
US_AI×NEWS_INJECTED	-3.2136***	-0.0157***
	(0.062)	(0.003)
US_AI	2.4151***	0.0135***
	(0.722)	(0.004)
NEWS_INJECTED	0.0023	0.0000
	(0.002)	(0.000)
FIRM_SIZE	0.6884	-0.0036
	(0.797)	(0.003)
ROA	33.6489***	-0.0279
	(7.811)	(0.061)
LEVERAGE	-16.2887***	-0.0243***
	(3.688)	(0.006)
LOSS	0.5618	-0.0123
	(1.025)	(0.013)
BM	-4.1208***	-0.0475***
	(1.213)	(0.012)
PRE_RETURN	-8.8492***	-0.0015
	(0.968)	(0.009)
RETURN_VOLATILITY	5.0622***	-0.0011
	(0.797)	(0.002)
IO	0.0727*	0.0002
	(0.034)	(0.000)
Industry FE	Yes	Yes
Observations	19,912	19,912
Adj R^2	0.3087	0.0418

Table 7. AI Predictions of US firms

This table repeats the baseline analysis for a sample of 11,688 U.S. firm-year observations. The dependent variables are price predictions (Column 1) and directional forecasts (Column 2) for U.S. firms, with *US AI* indicating ChatGPT predictions. The coefficients on *US AI* are small and statistically insignificant, confirming that the documented foreign bias is specific to cross-border predictions about Chinese firms. All continuous variables are winsorized at the 1st and 99th percentiles. All variables are defined in Appendix A. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1)	(2)
	AI_PREDICTION_US	*AI_RECOMMEND_US*
US_AI	-0.1539	-0.0472
	(0.361)	(0.027)
FIRM_SIZE	10.8241***	0.0191***
	(2.254)	(0.004)
ROA	-3.2749***	0.0183***
	(0.674)	(0.004)
LEVERAGE	-0.0487	-0.0030
	(0.183)	(0.003)
LOSS	-31.9823***	-0.0500**
	(5.347)	(0.022)
BM	-4.6968***	-0.0017
	(0.931)	(0.004)
PRE_RETURN	0.7273***	0.0061*
	(0.184)	(0.003)
RETURN_VOLATILITY	-0.0857	-0.0099**
	(0.087)	(0.004)
IO	12.9281	-0.0110
	(11.715)	(0.013)
Industry FE	Yes	Yes
Observations	11,688	11,688
Adj R^2	0.2740	0.1759

Table 8. Synthetic Firms

This table investigates whether foreign bias persists when both U.S. and Chinese media coverage is absent. We reports results for a placebo set of synthetic Chinese firm names. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

	(1) *AI_PREDICTION*	(2) *AI_RECOMMEND*
US_AI	-0.2260	0.0100
	(0.275)	(0.029)
FIRM_SIZE	-0.5009	-0.0379
	(0.801)	(0.023)
ROA	-47.7204*	-0.2994
	(22.845)	(0.423)
LEVERAGE	0.3245	0.0444
	(6.373)	(0.108)
LOSS	-9.1091**	-0.1418*
	(3.396)	(0.066)
BM	0.1858	0.0306
	(1.098)	(0.054)
PRE_RETURN	-4.4750	-0.0586
	(3.371)	(0.085)
RETURN_VOLATILITY	2.5929**	0.0001
	(0.863)	(0.037)
IO	0.0567	0.0016
	(0.036)	(0.001)
Industry FE	Yes	Yes
Observations	200	200
Adj R^2	0.1773	0.0909

Table 9. Testing the Elicitation Channel

This table presents two robustness analyses to test the hypothesis that the foreign bias is a result of differences in how each LLM responds to prompts. Panel A repeats the baseline prompts in Mandarin. Panel B enables web search for both models before prediction. Panel C repeats the baseline prompts using ChatGPT-5-Nano API. As ChatGPT-5 cannot set the temperature to 0, we took the average (median) of 50 random draws to measure the US AI's stock price prediction (directional forecasts). Panel D extends the prediction horizon to 12 months. All continuous variables are winsorized at the 1st and 99th percentiles. All variables are defined in Appendix A and all continuous variables are winsorized at the 1st and 99th percentiles. Standard errors, clustered at the Fama-French 12-industry level, are reported in parentheses. Significance levels are indicated by *, **, and *** for 10%, 5%, and 1% respectively.

Panel A: AI Prediction with Prompt in Mandarin

	(1)	(2)
	AI_PREDICTION_Mandarin	*AI_RECOMMEND_Mandarin*
US_AI	1.5173**	0.0135*
	(0.644)	(0.007)
FIRM_SIZE	3.8150	0.0157***
	(2.549)	(0.004)
ROA	60.9911**	0.0607
	(23.239)	(0.077)
LEVERAGE	-22.6838**	-0.0342*
	(7.758)	(0.017)
LOSS	5.5226	-0.0068
	(3.226)	(0.019)
BM	-7.3221	0.0498***
	(4.620)	(0.016)
PRE_RETURN	-7.4429***	0.0116
	(1.305)	(0.008)
RETURN_VOLATILITY	4.0061***	0.0019
	(0.928)	(0.004)
IO	0.0553	-0.0003**
	(0.039)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.0763	0.0175

Table 9 [Continued]

Panel B: AI Prediction Allowing Online Search

	(1)	(2)
	AI_PREDICTION_Web	*AI_RECOMMEND_Web*
US_AI	0.5338**	0.0426**
	(0.228)	(0.016)
FIRM_SIZE	0.9237***	0.0072
	(0.151)	(0.013)
ROA	2.0662	-1.2009***
	(2.513)	(0.212)
LEVERAGE	-0.4815	0.3108***
	(0.455)	(0.067)
LOSS	0.9130***	0.0068
	(0.249)	(0.021)
BM	-1.3104***	0.0653**
	(0.321)	(0.024)
PRE_RETURN	0.2101	0.1196***
	(0.357)	(0.029)
RETURN_VOLATILITY	-0.5043**	-0.1436***
	(0.164)	(0.021)
IO	-0.0077*	-0.0015*
	(0.004)	(0.001)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.0186	0.2017

Table 9 [Continued]

Panel C: AI Prediction Using ChatGPT-5

	(1)	(2)
	AI_PREDICTION_ChatGPT5	*AI_RECOMMEND_ChatGPT5*
US_AI	1.6622**	0.0221***
	(0.704)	(0.005)
FIRM_SIZE	3.5656	-0.0025
	(2.135)	(0.001)
ROA	53.6364**	-0.0031
	(19.005)	(0.031)
LEVERAGE	-21.8180***	-0.0094**
	(6.630)	(0.004)
LOSS	3.9178	-0.0015
	(2.679)	(0.006)
BM	-8.6852**	-0.0332***
	(3.801)	(0.010)
PRE_RETURN	-7.4982***	-0.0031
	(0.958)	(0.005)
RETURN_VOLATILITY	4.1449***	-0.0002
	(0.820)	(0.001)
IO	0.0502	0.0001
	(0.030)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.0823	0.0395

Table 9 [Continued]

Panel D: Alternative Prediction Horizon

	(1)	(2)
	AI_PREDICTION_ALT	*AI_RECOMMEND_ALT*
US_AI	1.8490**	0.0207***
	(0.818)	(0.006)
FIRM_SIZE	4.2758	-0.0059
	(2.668)	(0.005)
ROA	71.6423**	-0.0357
	(24.110)	(0.094)
LEVERAGE	-26.2111***	-0.0284**
	(8.427)	(0.012)
LOSS	6.1597	-0.0283
	(3.553)	(0.020)
BM	-10.1748*	-0.0453**
	(4.901)	(0.016)
PRE_RETURN	-9.3972***	-0.0084
	(1.374)	(0.011)
RETURN_VOLATILITY	4.8821***	0.0041
	(1.037)	(0.003)
IO	0.0624	0.0000
	(0.039)	(0.000)
Industry FE	Yes	Yes
Observations	9,956	9,956
Adj R^2	0.1037	0.0479